

Project Overview

This project automates the **archiving of old S3 files** using an **EC2 Shell Script + AWS CLI**, then uploads compressed archives back to S3 and sends an **SNS alert**.

FINAL OUTPUT

- ✓ Automatically checks S3 bucket for files older than X days
 - ✓ Downloads → compresses → uploads to /archive/ folder
 - ✓ Sends SNS alert: **“Archiving Completed”**
 - ✓ Runs daily using **cron job**
-

Step-by-Step Implementation

STEP 1 — Create S3 Buckets

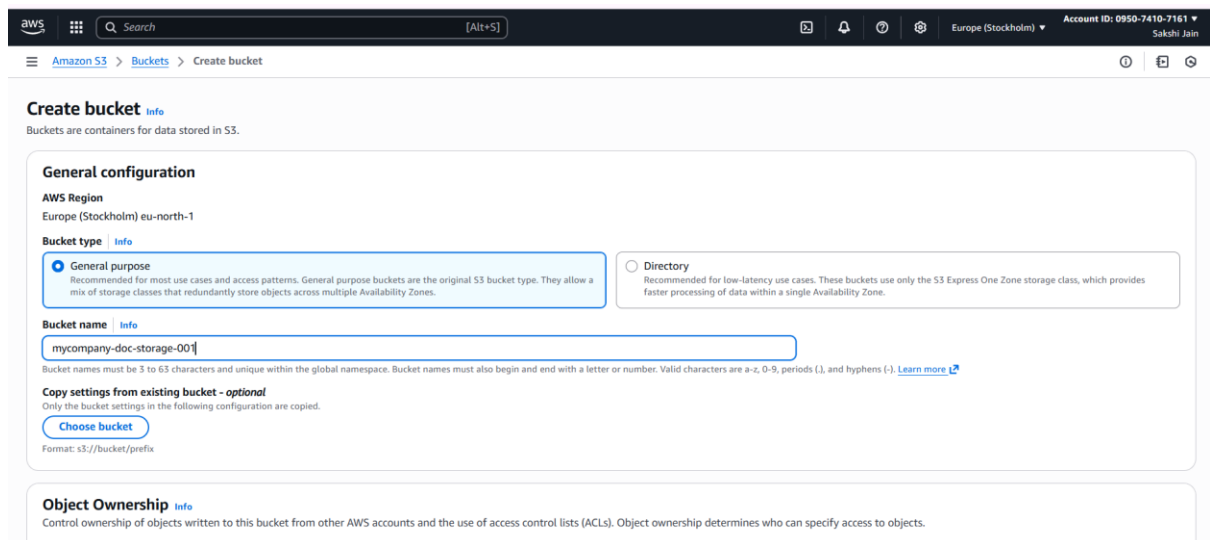
1. Create a main bucket

Example:

mycompany-doc-storage-001

2. Inside it, create a folder:

archive/



The screenshot shows the AWS Management Console interface for creating a new S3 bucket. The top navigation bar includes the AWS logo, a search bar, and account information for 'Europe (Stockholm)' with account ID '0950-7410-7161'. The breadcrumb trail indicates the path: Amazon S3 > Buckets > Create bucket. The main heading is 'Create bucket' with an 'info' link. Below this, a sub-header 'General configuration' is followed by the 'AWS Region' set to 'Europe (Stockholm) eu-north-1'. The 'Bucket type' section offers two options: 'General purpose' (selected with a radio button) and 'Directory'. The 'General purpose' option is described as recommended for most use cases and access patterns, allowing a mix of storage classes. The 'Directory' option is recommended for low-latency use cases using the S3 Express One Zone storage class. The 'Bucket name' field contains 'mycompany-doc-storage-001'. A note specifies that bucket names must be 3 to 63 characters and unique within the global namespace. A 'Choose bucket' button is located below the name field. The 'Object Ownership' section is partially visible at the bottom, mentioning control ownership and access control lists (ACLs).

Create folder [Info](#)

Use folders to group objects in buckets. When you create a folder, S3 creates an object using the name that you specify followed by a slash (/). This object then appears as folder on the console. [Learn more](#)

Your bucket policy might block folder creation

If your bucket policy prevents uploading objects without specific tags, metadata, or access control list (ACL) grantees, you will not be able to create a folder using this configuration. Instead, you can use the [upload configuration](#) to upload an empty folder and specify the appropriate settings.

Folder

Folder name

Folder names can't contain */*. [See rules for naming](#)

Server-side encryption [Info](#)

Server-side encryption protects data at rest.

The following encryption settings apply only to the folder object and not to sub-folder objects.

Server-side encryption

☒ Don't specify an encryption key
The bucket settings for default encryption are used to encrypt the folder object when storing it in Amazon S3.

☐ Specify an encryption key

STEP 2 — Create SNS Topic for Alerts

Create Topic:

S3-Archive-Alerts

New Feature
Amazon SNS now supports High Throughput FIFO topics. [Learn more](#)

Application Integration

Amazon Simple Notification Service

Pub/sub messaging for microservices and serverless applications.

Amazon SNS is a highly available, durable, secure, fully managed pub/sub messaging service that enables you to decouple microservices, distributed systems, and event-driven serverless applications. Amazon SNS provides topics for high-throughput, push-based, many-to-many messaging.

Create topic

Topic name

A topic is a message channel. When you publish a message to a topic, it fans out the message to all subscribed endpoints.

Next step

[Start with an overview](#)

Pricing

Subscribe Email:

Your email → Confirm from inbox.

SNS Topic ARN will look like:

arn:aws:sns:eu-north-1:095074107161:S3-Archive-Alerts

AWS Search [Alt+S] Europe (Stockholm) Account ID: 0950-7410-7161 Sakshi Jain

Amazon SNS > Subscriptions > Create subscription

New Feature
Amazon SNS now supports High Throughput FIFO topics. [Learn more](#)

Create subscription

Details

Topic ARN
arn:aws:sns:eu-north-1:095074107161:S3-Archive-Alerts

Protocol
The type of endpoint to subscribe
Email

Endpoint
An email address that can receive notifications from Amazon SNS.
sakshimiss0@gmail.com

After your subscription is created, you must confirm it. [Info](#)

Subscription filter policy - optional [Info](#)
This policy filters the messages that a subscriber receives.

AWS Notifications <no-reply@sns.amazonaws.com> 10:34 AM (3 minutes ago) ☆ 😊 ↩ ⋮
to me ▾

Why is this message in spam? This message is similar to messages that were identified as spam in the past.

Report not spam ⓘ

You have chosen to subscribe to the topic:
arn:aws:sns:eu-north-1:095074107161:S3-Archive-Alerts

To confirm this subscription, click or visit the link below (If this was in error no action is necessary):
[Confirm subscription](#)

Please do not reply directly to this email. If you wish to remove yourself from receiving all future SNS subscription confirmation requests please send an email to [sns-opt-out](#)



Simple Notification Service

Subscription confirmed!

You have successfully subscribed.

Your subscription's id is:
arn:aws:sns:eu-north-1:095074107161:S3-Archive-Alerts:f8cf5af5-6bf8-42e5-a36f-7fd92c598bdb

If it was not your intention to subscribe, [click here to unsubscribe](#).

STEP 3 — Create IAM Role for EC2

Role Name:

EC2-S3Archive-Role

Attach Policies:

✓ AmazonSNSFullAccess

aws

[Alt+S]

Search

Global

Account ID: 0950-7410-7161

Sakshi Jain

IAM > Roles > Create role

Step 1
Select trusted entity
Step 2
Add permissions
Step 3
Name, review, and create

Select trusted entity info

Trusted entity type

☒ AWS service
Allow AWS services like EC2, Lambda, or others to perform actions in this account.

☐ AWS account
Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.

☐ Web identity
Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.

☐ SAML 2.0 federation
Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.

☐ Custom trust policy
Create a custom trust policy to enable others to perform actions in this account.

Use case

Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case
EC2

Choose a use case for the specified service.
Use case
EC2

Choose one or more policies to attach to your new role.

Add permissions

Permissions policies (2/1084)

Choose one or more policies to attach to your new role.

Filter by Type: All types 5 matches

Policy name	Type	Description
☒ AmazonSNSFullAccess	AWS managed	Provides full access to Amazon SNS via...

aws

Search

[Alt+S]

Global

Account ID: 0950-7410-7161

Sakshi Jain

IAM

>

Roles

>

Create role

Step 1

Select trusted entity

Step 2

Add permissions

Step 3

Name, review, and create

Name, review, and create

Role details

Role name

Enter a meaningful name to identify this role.

EC2-S3Archive-Role

Maximum 64 characters. Use alphanumeric and `^_~@-.` characters.

Description

Add a short explanation for this role.

Allows EC2 instances to call AWS services on your behalf.

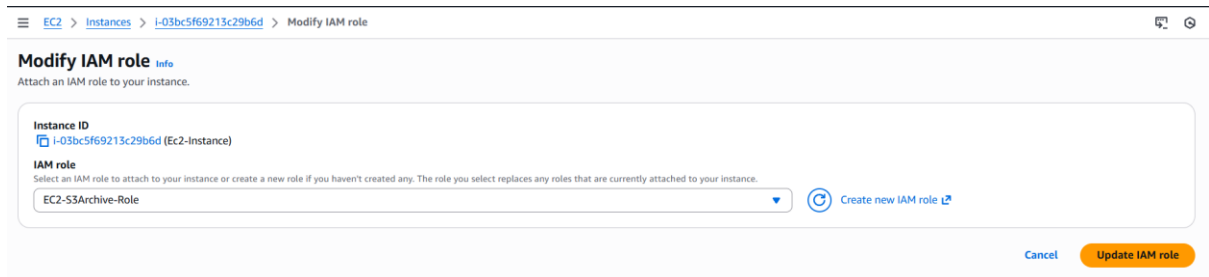
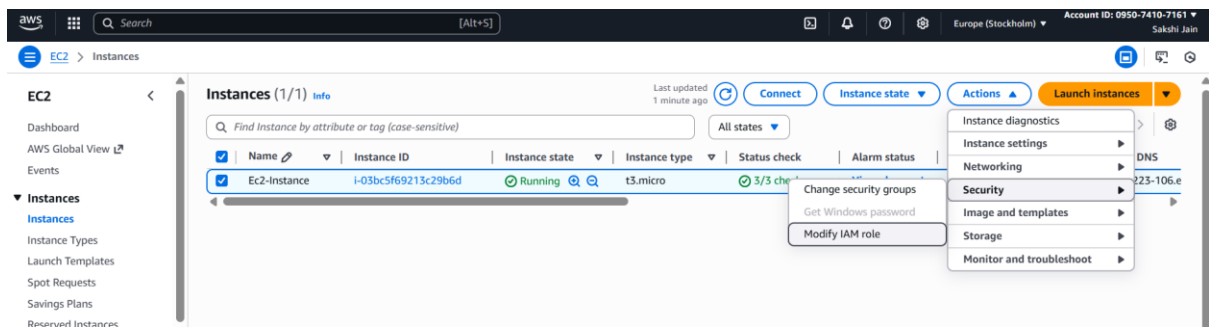
Maximum 1000 characters. Use letters (A-Z and a-z), numbers (0-9), tabs, new lines, or any of the following characters: `_~@-./(){}$%'^`:"'``.

Step 1: Select trusted entities

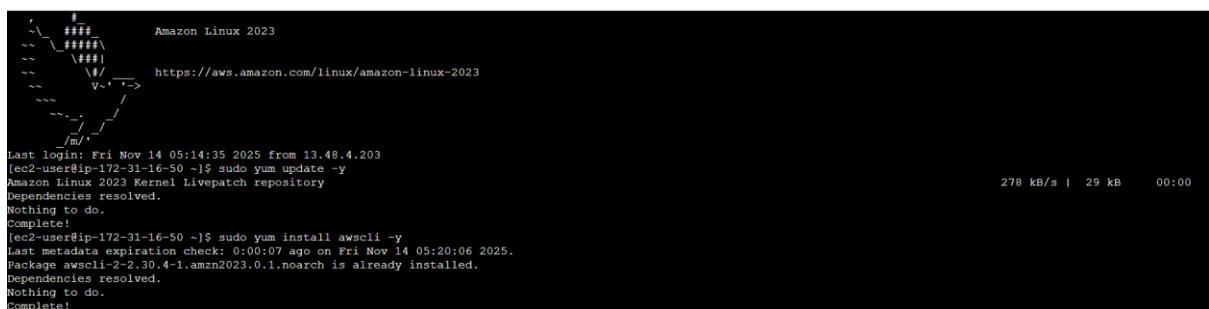
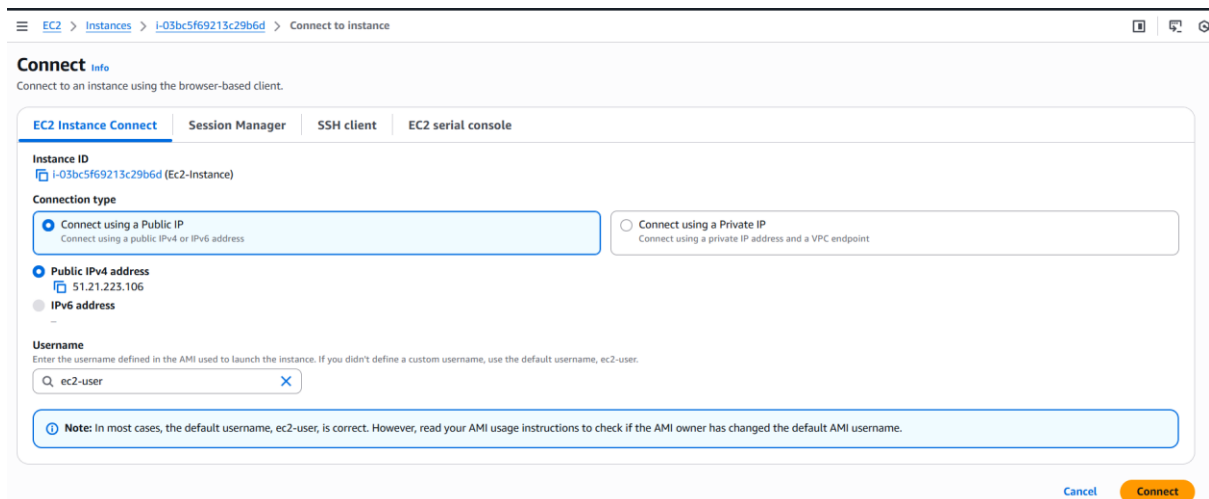
Trust policy

```
1- {
2-   "Version": "2012-10-17",
3-   "Statement": [
4-     {
5-       "Effect": "Allow",
6-       "Action": [
```

Attach IAM Role



Connect with EC2 Instance



STEP 5 — Create Archive Script on EC2

Create the script:

```
vim /home/ec2-user/s3_archive.sh
```

Paste the script:

```
#!/bin/bash
```

```
BUCKET=" mycompany-doc-storage-001"
```

```
ARCHIVE_FOLDER="archive"
```

```
DAYS_OLD=0
```

```
SNS_TOPIC="arn:aws:sns:eu-north-1:095074107161:S3-Archive-Alerts"
```

```
WORKDIR="/home/ec2-user/archive_work"
```

```
mkdir -p $WORKDIR
```

```
echo "Starting S3 Archive Process..."
```

```
# Get list of files older than X days
```

```
aws s3api list-objects --bucket $BUCKET --query \
```

```
"Contents[?LastModified<=`date -d \"$DAYS_OLD days ago\" +%Y-%m-%dT%H:%M:%S`].{Key:Key}\" \
```

```
--output text > $WORKDIR/files.txt
```

```
if [ ! -s $WORKDIR/files.txt ]; then
```

```
    echo "No files older than $DAYS_OLD days."
```

```
    exit 0
```

```
fi
```

```
cd $WORKDIR
```

```
# Process each file
```

```
while read FILE; do
```

```
    echo "Processing: $FILE"
```

```
# Download file
```

```
aws s3 cp s3://$BUCKET/$FILE .
```

```
done < files.txt
```

```
# Create compressed file

ARCHIVE_NAME="archive_$(date +%F).tar.gz"

tar -czf $ARCHIVE_NAME *


# Upload to S3 archive folder

aws s3 cp $ARCHIVE_NAME s3://$BUCKET/$ARCHIVE_FOLDER/


# Send SNS Notification

aws sns publish \

    --topic-arn "$SNS_TOPIC" \

    --message "S3 Archiving Completed. Files archived: $ARCHIVE_NAME"


echo "Archiving Done!"
```



The image shows two overlapping windows. The top window is a terminal with the command `[ec2-user@ip-172-31-16-50 ~]$ vim /home/ec2-user/s3_archive.sh`. The bottom window is the AWS Management Console, displaying the content of the `s3_archive.sh` script. The script content is as follows:

```
exit 0

fi

cd $WORKDIR

# Process each file
while read FILE; do
    echo "Processing: $FILE"

    # Download file
    aws s3 cp s3://$BUCKET/$FILE .

done < files.txt

# Create compressed file
ARCHIVE_NAME="archive_$(date +%F).tar.gz"
tar -czf $ARCHIVE_NAME *

# Upload to S3 archive folder
aws s3 cp $ARCHIVE_NAME s3://$BUCKET/$ARCHIVE_FOLDER/

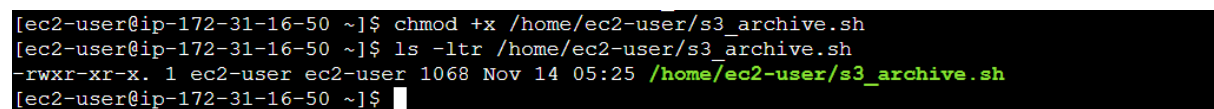
# Send SNS Notification
aws sns publish \
    --topic-arn "$SNS_TOPIC" \
    --message "S3 Archiving Completed. Files archived: $ARCHIVE_NAME"

echo "Archiving Done!"

-- INSERT --
```

STEP 6 — Make Script Executable

```
chmod +x /home/ec2-user/s3_archive.sh
```



The image shows a terminal window with the following commands and output:

```
[ec2-user@ip-172-31-16-50 ~]$ chmod +x /home/ec2-user/s3_archive.sh
[ec2-user@ip-172-31-16-50 ~]$ ls -ltr /home/ec2-user/s3_archive.sh
-rwxr-xr-x. 1 ec2-user ec2-user 1068 Nov 14 05:25 /home/ec2-user/s3_archive.sh
[ec2-user@ip-172-31-16-50 ~]$
```

STEP 7 — Test Script Manually

./s3_archive.sh

You should see:

- ✓ Files downloaded
- ✓ Archive created
- ✓ Uploaded to /archive/ folder
- ✓ Email alert received from SNS

```
Processing: umesh_accessKeys.csv
download: s3://my-company-doc-storage/umesh_accessKeys.csv to ./umesh_accessKeys.csv
Processing: umesh_credentials.csv
download: s3://my-company-doc-storage/umesh_credentials.csv to ./umesh_credentials.csv
tar: archive_2025-11-14.tar.gz: File shrank by 388 bytes; padding with zeros
upload: ./archive_2025-11-14.tar.gz to s3://my-company-doc-storage/archive/archive_2025-11-14.tar.gz
{
  "MessageId": "bbf9fad0-8f17-558a-8d90-9b4c4b830dec"
}
Archiving Done!
[ec2-user@ip-172-31-20-216 ~]$ ./s3_archive.sh
Starting S3 Archive Process...
Processing: archive/
download failed: s3://my-company-doc-storage/archive/ to ./ [Errno 20] Not a directory: '/home/ec2-user/archive_work/.FC8Cbeb7' -> '/home/ec2-user/archive_work/'
Processing: archive/archive_2025-11-14.tar.gz
download: s3://my-company-doc-storage/archive/archive_2025-11-14.tar.gz to ./archive_2025-11-14.tar.gz
Processing: umesh_accessKeys.csv
download: s3://my-company-doc-storage/umesh_accessKeys.csv to ./umesh_accessKeys.csv
Processing: umesh_credentials.csv
download: s3://my-company-doc-storage/umesh_credentials.csv to ./umesh_credentials.csv
tar: archive_2025-11-14.tar.gz: File shrank by 431 bytes; padding with zeros
upload: ./archive_2025-11-14.tar.gz to s3://my-company-doc-storage/archive/archive_2025-11-14.tar.gz
{
  "MessageId": "3a551d69-ae0e-53a3-9457-7cf969b4030b"
}
Archiving Done!
[ec2-user@ip-172-31-20-216 ~]$
```

archive/ Copy S3 URI

Objects Properties

Objects (1) Copy S3 URI Copy URL Download Open Delete Actions Create folder Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 Inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

☐	Name	Type	Last modified	Size	Storage class
☐	archive_2025-11-14.tar.gz	gz	November 14, 2025, 11:01:57 (UTC+05:30)	388.0 B	Standard

Gmail Search mail

1 of 4,790

AWS Notification Message Inbox

AWS Notifications <no-reply@sns.amazonaws.com> 11:13 AM (3 minutes ago) ☆ 😊 ↶ ⋮

S3 Archiving Completed. Files archived: archive_2025-11-14.tar.gz

--

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<https://sns.us-east-1.amazonaws.com/unsubscribe.html?SubscriptionArn=arn:aws:sns:us-east-1:930832106341:S3-Archive-Alertss:a97d8164-c9c3-4c5e-82f1-ce7e600afcf0&Endpoint=jay951054@gmail.com>

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